Motion Detection Using Arduino

Introduction

A Motion Sensor Security System is an electronic system designed to detect the movement of a person or object within a specific area. In this project, an Arduino Uno is used as the main controller, and a PIR (Passive Infrared) sensor is used to detect motion.

When the PIR sensor detects movement, the Arduino processes the sensor signal and activates an LED and buzzer as an alarm. When no motion is detected, the LED and buzzer remain OFF.

The complete circuit is designed and tested using the Tinkercad Circuits simulator, which allows the circuit and Arduino program to be tested virtually without physical hardware.

Objectives

The main objectives of this project are:

- To understand the working principle of a PIR motion sensor.
- To interface a PIR sensor with an Arduino Uno.
- To control an LED and buzzer using Arduino.
- To develop a simple security alarm system.
- To simulate and test the circuit using Tinkercad.
- To understand basic Arduino programming and digital input/output.

Components Required

S.No.	Component	Quantity
1	Arduino Uno R3	1
2	PIR Motion Sensor	1
3	LED	1
4	220 Ω Resistors	1
5	Piezo Buzzer	1
6	Breadboard	1
7	Jumper Wires	As Required

Working Principle

- The PIR sensor detects changes in infrared radiation caused by the movement of people or other objects.

- The PIR sensor has three main connections:

  - VCC → Arduino 5V
  - OUT/Signal → Arduino Digital Pin 2
  - GND → Arduino GND

- The Arduino continuously reads the output of the PIR sensor.

- When there is no motion, the PIR output is LOW. The Arduino keeps the LED and buzzer OFF.

- When motion is detected, the PIR output becomes HIGH. The Arduino detects this HIGH signal and switches ON:

  - The LED is connected to Digital Pin 13.
  - The buzzer is connected to Digital Pin 8.

- Therefore, the LED and buzzer provide a visual and audible indication that motion has been detected.

Circuit Connections

- **PIR Sensor**

  PIR Sensor| Arduino Uno
  VCC| 5V
  OUT/Signal| Digital Pin 2
  GND| GND

- **LED**

  The LED is connected through a 220 Ω resistor.

  Arduino D13 → 220 Ω resistors → LED positive/long leg

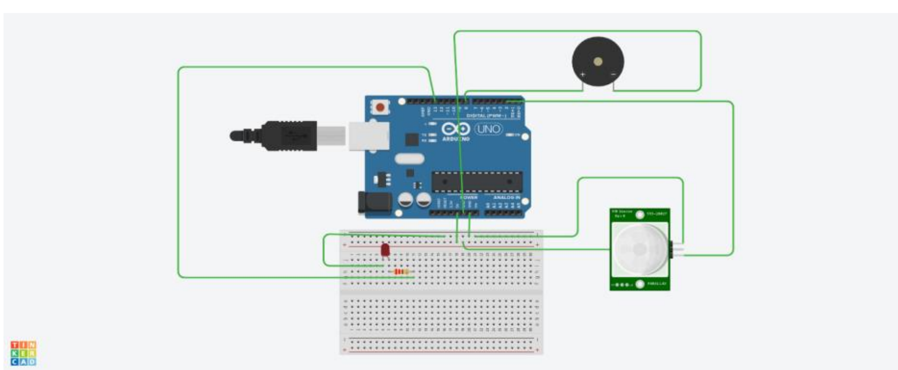
  LED negative/short leg → GND

- **Buzzer**

  Buzzer positive (+) → Arduino D8

  Buzzer negative (−) → GND

  https://www.tinkercad.com/things/2OU6JmQM4hj-motion-detection?sharecode=yc53vw0Sr5uJPjFYk0tH3lpNIztFXWGBJLCGgIE8bBQ



Arduino Program

```
#define PIR_PIN 2
#define LED_PIN 13
#define BUZZER_PIN 8

void setup() {
  pinMode(PIR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);

  Serial.begin(9600);
}

void loop() {
  if (digitalRead(PIR_PIN) == HIGH) {

    digitalWrite(LED_PIN, HIGH);
    digitalWrite(BUZZER_PIN, HIGH);

    Serial.println("MOTION DETECTED!");

  } else {

    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);

    Serial.println("NO MOTION");
  }

  delay(500);
}
```

Software Used

Tinkercad Circuits

Tinkercad Circuits is used to create and simulate the Arduino circuit virtually. It allows the components to be connected, the Arduino program to be uploaded, and the circuit operation to be tested without requiring physical components.

Arduino Programming

The Arduino program is written using C/C++-based Arduino programming syntax. The program reads the PIR sensor and controls the LED and buzzer according to the detected motion.

Advantages

- Simple and inexpensive system.
- Easy to build and program.
- Provides both visual and audio alerts.
- PIR sensors consume relatively low power.
- Can be used for basic security applications.
- Easy to expand with additional components.

Applications

The system can be used for:

- Home security.
- Room intrusion detection.
- Office security.
- Laboratory monitoring.
- Door and entrance monitoring.
- Storage-room security.
- Automatic lighting systems.

Limitations

- The PIR sensor detects motion based on changes in infrared radiation.
- It may not detect stationary objects.
- Detection range depends on the PIR sensor.
- Environmental conditions can affect sensor performance.
- The basic project does not provide remote notifications.

Result

The Motion Sensor Security System was successfully designed and simulated using Arduino Uno, PIR sensor, LED, and buzzer in Tinkercad.

When the PIR sensor detects motion, the Arduino receives a HIGH signal and activates the LED and buzzer. When motion is not detected, the Arduino switches them OFF.

Thus, the project successfully demonstrates the use of a PIR sensor with Arduino for basic motion detection and security applications.

Conclusion

The Motion Sensor Security System provides a simple and effective method for detecting movement. Through this project, the working of a PIR sensor, Arduino digital input/output, LED control, and buzzer control were studied.

The use of Tinkercad made it possible to design and test the circuit virtually before implementing it with physical hardware. The project can serve as a foundation for developing more advanced IoT-based security systems.